

# Report of Findings

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## Abstract

### Viability of Existing CSR Dataset for Generative RAG LLM Readiness

The analysis herein provides a structured, multi-level evaluation of the Confluence-based Customer Service Representative (CSR) knowledge repository, ensuring its readiness for ingestion into GCP Vertex AI to support a Retrieval-Augmented Generation (RAG) Large Language Model (LLM). This approach effectively balances quantitative assessments of data composition and quality with qualitative insights into linguistic complexity, knowledge coverage, and retrieval efficiency—key factors influencing the effectiveness of a generative AI system.

- **Dimensionality** examines the structural integrity and completeness of the data, ensuring it is statistically robust and sufficiently populated for downstream AI applications.
- **Microanalysis** focuses on token distribution, frequency, and correlation, which are essential for determining optimal chunking strategies and assessing retrieval granularity.

- **Mesoanalysis** extends this by measuring lexical variety and richness, offering insight into data diversity, redundancy, and uniqueness—critical elements for ensuring meaningful retrieval and generation.
- **Macroanalysis** leverages clustering, classification, and topic modeling to extract emergent themes and structural patterns, which directly inform indexing strategies, embedding generation, and retrieval optimization.
- **Takeaways** consolidate findings to provide actionable recommendations for refining the preprocessing pipeline, optimizing searchability, and enhancing model alignment with CSR workflows.

This structured assessment ensures that the [REDACTED] CSR knowledge base is not only ingestible by GCP Vertex AI but also efficiently retrievable, leading to highly contextual and accurate LLM responses. The methodology aligns with best practices for retrieval-enhanced generative AI, validating that the data is well-prepared for embedding storage, retrieval strategies, and fine-tuning, ultimately enabling a scalable and high-performing RAG system for customer support automation.

We begin with loading the data from the Confluence API and the necessary mechanics around libraries involved:

```
## Rows: 179 Columns: 4
## — Column specification
##
## Delimiter: ","
## chr (4): Space, Title, URL, Content
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
## # A tibble: 6 × 4
##   Space Title URL Content
##   <chr> <chr> <chr> <chr>
## 1 ACM AC Manual https://[REDACTED]sound.atlassian... ACCOU
N...
## 2 ACM General Duties https://[REDACTED]sound.atlassian... Accou
n...
## 3 ACM Texting With Customers (SMS) https://[REDACTED]sound.atlassian... If yo
u...
## 4 ACM Office Hours https://[REDACTED]sound.atlassian... Vendo
r...
## 5 ACM Phone Calls https://[REDACTED]sound.atlassian... Rollo
v...
## 6 ACM Business Plans https://[REDACTED]sound.atlassian... Accou
n...
```

# Dimensionality

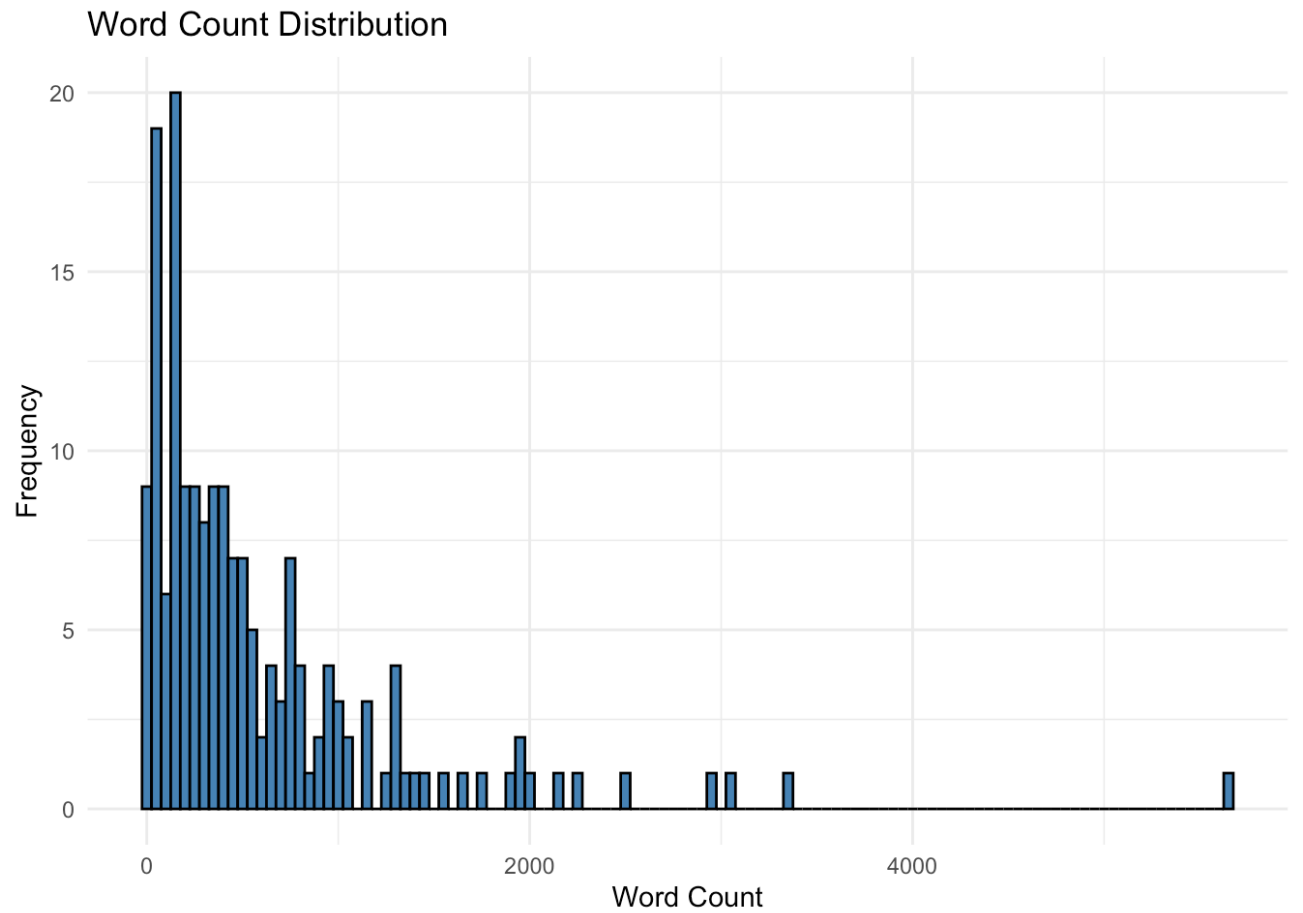
This section evaluates the structural integrity of the Confluence data, focusing on statistical composition and data completeness.

## Statistical Composition

The statistical composition of the Confluence repository provides insights into the overall structure and distribution of textual content, which is critical for evaluating its suitability for ingestion into a RAG-based LLM on GCP Vertex AI. This analysis quantifies key attributes such as the total number of pages, the word count distribution, and the variance in document length to determine whether the dataset is dense and information-rich or sparse and fragmented.

A dataset with consistently well-populated pages ensures that the AI model has sufficient context to generate meaningful responses, while an uneven word count distribution may indicate inconsistencies in documentation depth, with some pages being overly detailed and others lacking essential information. Additionally, extreme variations in document length may require preprocessing strategies such as chunking to optimize retrieval efficiency. By establishing these statistical baselines, we can determine whether further content normalization, augmentation, or filtering is necessary before deploying this knowledge base into a retrieval pipeline for generative AI-driven customer service support.

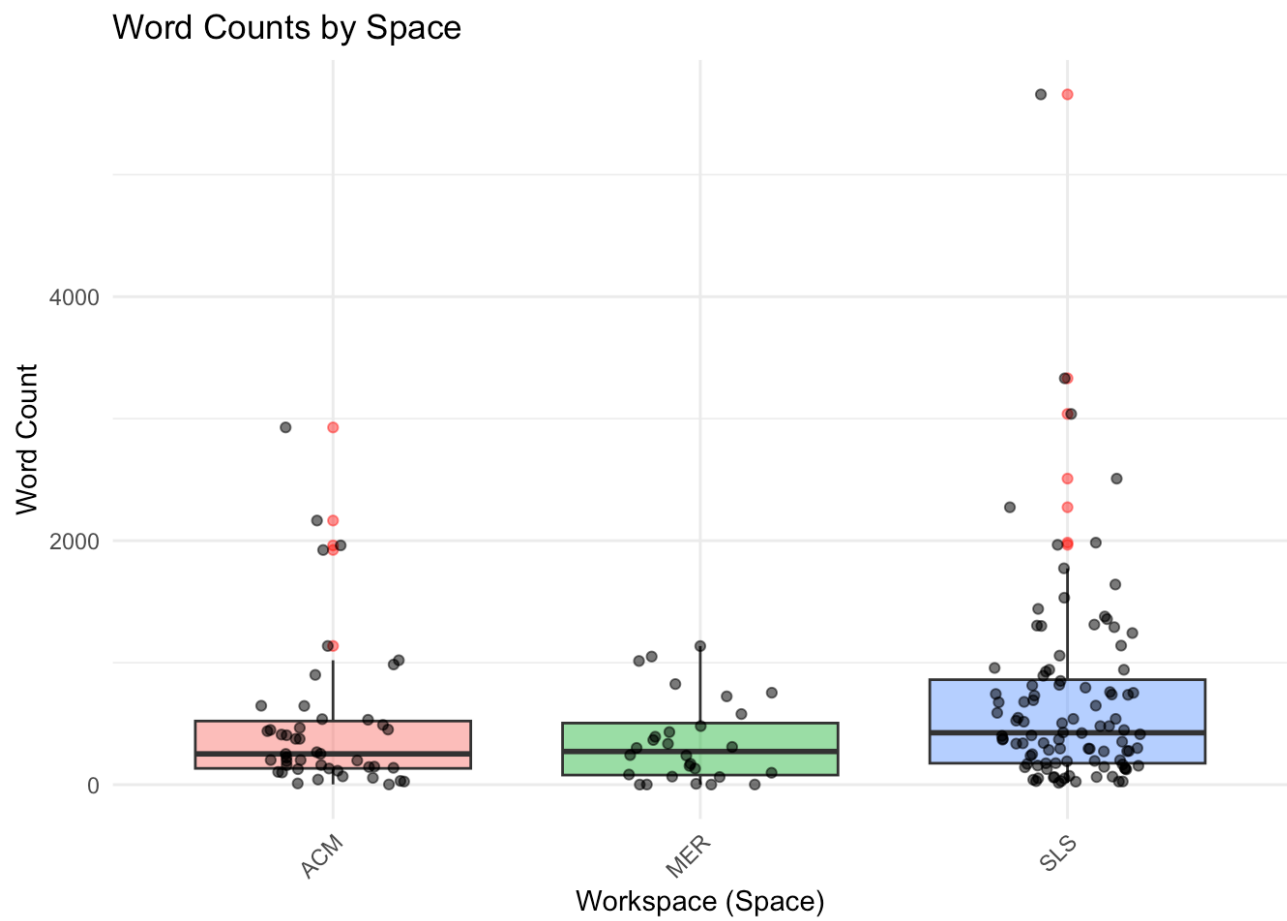
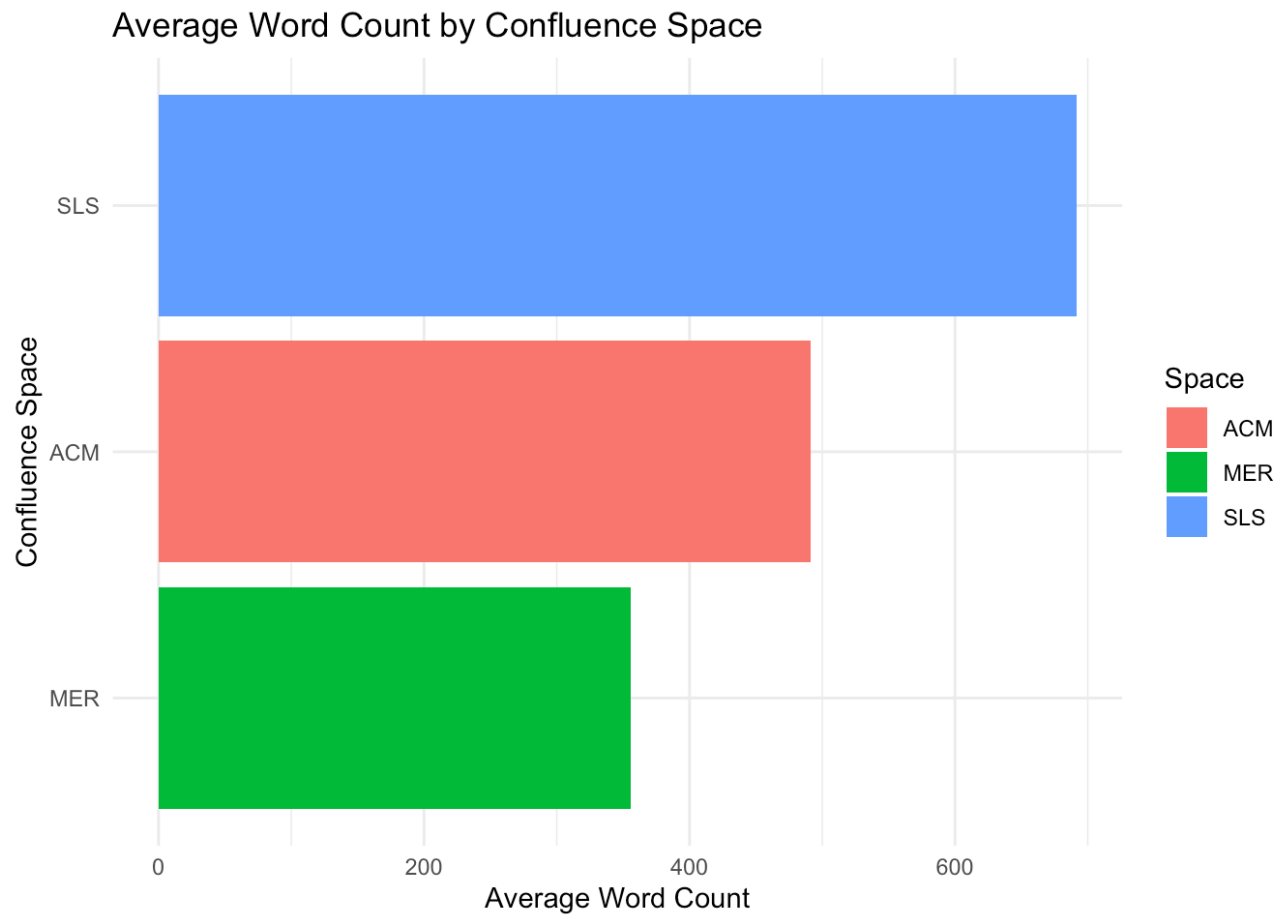
|                      |            |                 |          |          |
|----------------------|------------|-----------------|----------|----------|
| ## # A tibble: 1 × 4 |            |                 |          |          |
| ##                   | TotalPages | AvgWordsPerPage | MaxWords | MinWords |
| ##                   | <int>      | <dbl>           | <int>    | <int>    |
| ## 1                 | 179        | 584.            | 5658     | 1        |



```
## # A tibble: 1 × 10
##   TotalDocuments AvgWordsPerPage MedianWords MaxWords MinWords   Q1   Q3   I
##   <int>          <dbl>          <dbl>    <int>    <int> <dbl> <dbl> <dbl>
## 1      179          584.          370.    5658        1  146.  750.  604
## # i 2 more variables: LowerBound <dbl>, UpperBound <dbl>
```

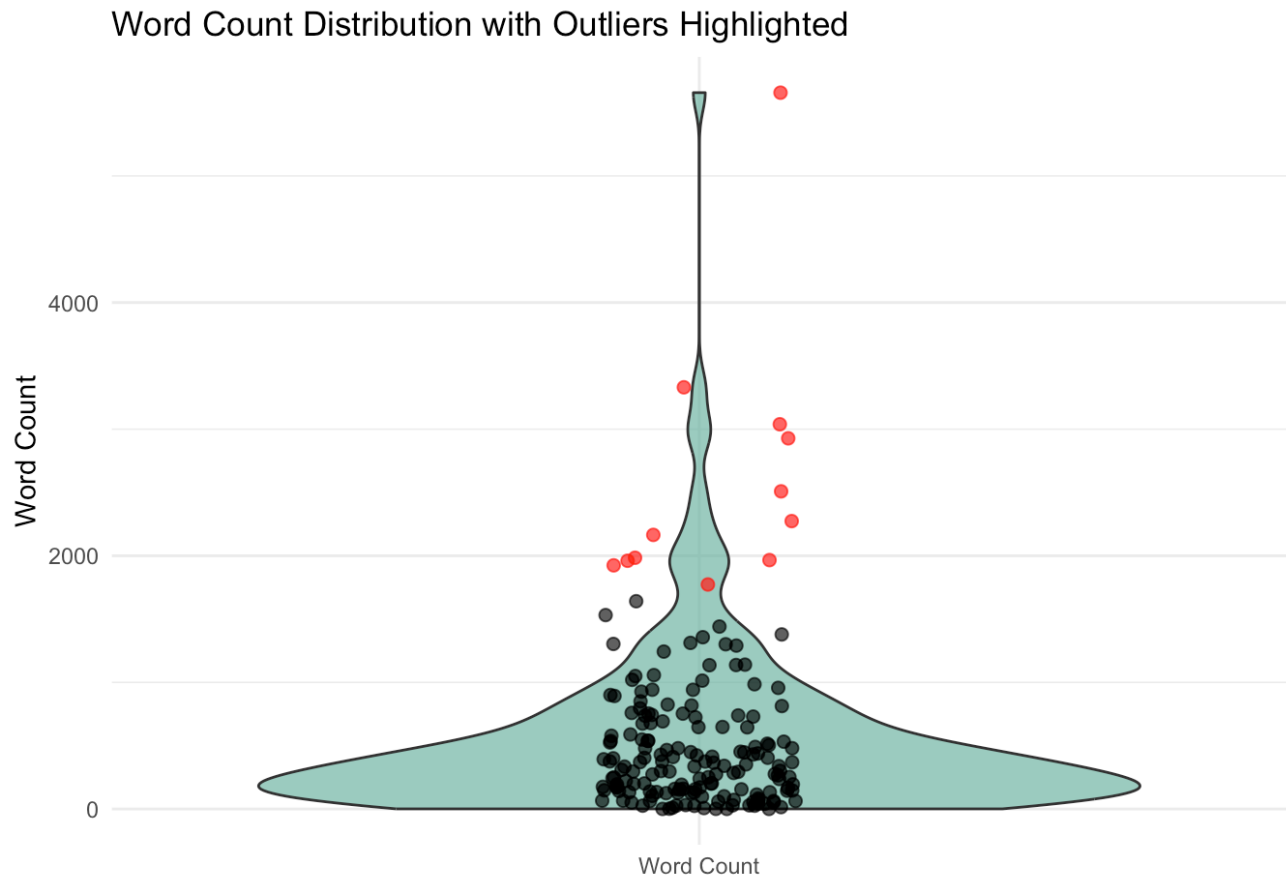
```
## # A tibble: 12 × 2
##   Title                               WordCount
##   <chr>                               <int>
## 1 Shipping                           2165
## 2 Message Checking                   2928
## 3 Snippet Examples                   1924
## 4 Mac keyboard shortcuts              1961
## 5 4D Remote and Outlook Mail Weblink 1773
## 6 ██████████ Card Application Processes 2274
## 7 Credit/Debit Information            3039
## 8 Names Simple Find                   3331
## 9 Names Finds                         5658
## 10 Names Menu Options                 1984
## 11 Invoice Menu Options                1966
## 12 Easy Pay                           2509
```

```
## # A tibble: 3 × 5
##   Space AvgWords MedianWords MaxWords MinWords
##   <chr>   <dbl>          <dbl>    <int>    <int>
## 1 ACM     491.          252.    2928        2
## 2 MER     356.          272    1136        1
## 3 SLS     691.          426.    5658       15
```



The following violin plot displays the distribution of document word counts within the Confluence dataset, providing a smoothed density visualization that highlights where most documents fall in terms of length. The width of the plot represents the frequency of documents at different word counts, giving a clearer picture of content variability compared to a traditional boxplot.

The red points indicate outliers, which are documents that fall significantly outside the expected range based on the  $1.5 \times \text{IQR}$  rule. These outliers may represent unusually short or excessively long documents, which could impact retrieval performance in a RAG-based AI system. By identifying these variations, we can assess whether content standardization is needed to improve the dataset's retrievability and response accuracy in GCP Vertex AI.

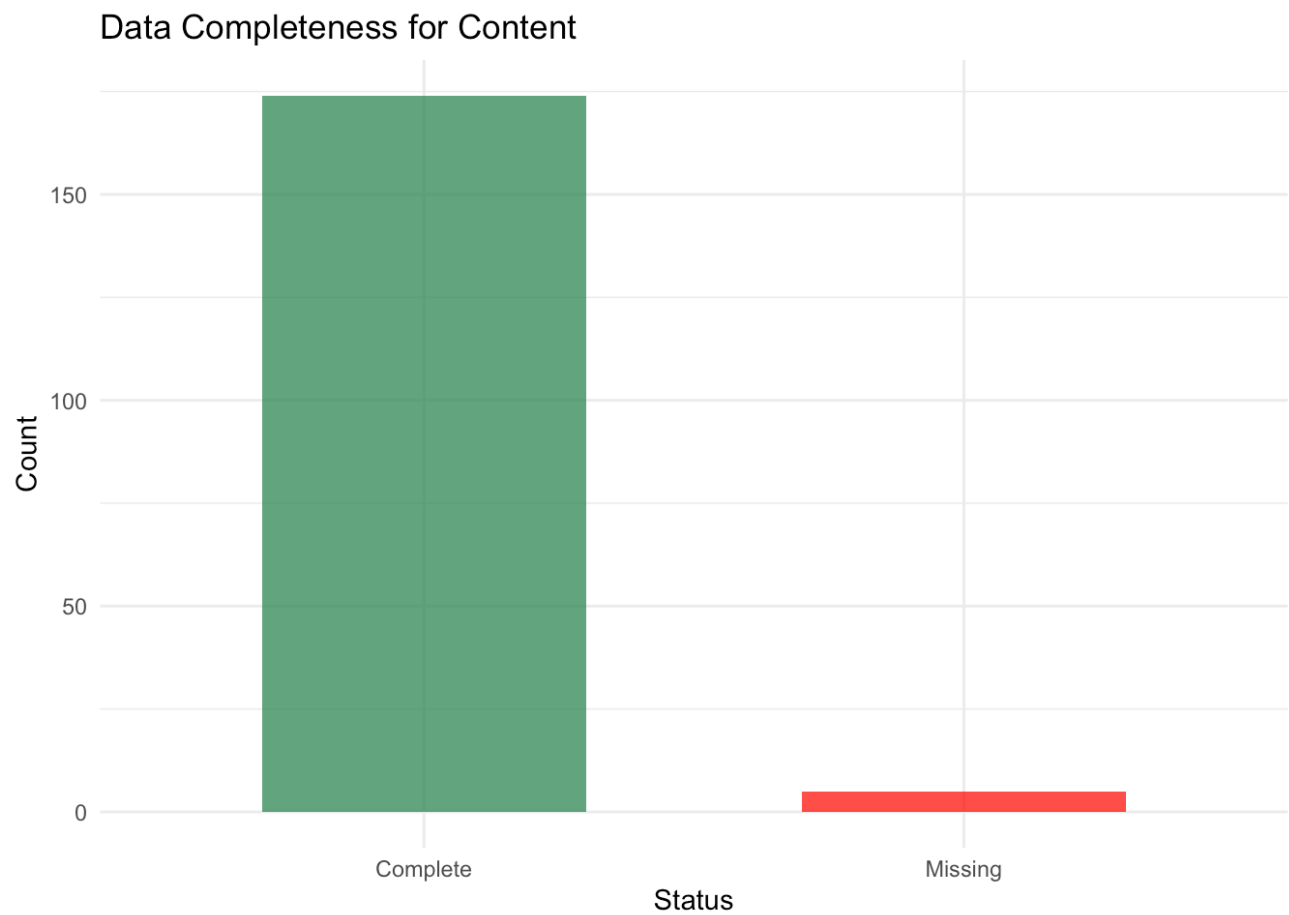


## Data Completeness

Data completeness is a fundamental aspect of assessing the readiness of the Confluence repository for ingestion into GCP Vertex AI to support a Retrieval-Augmented Generation (RAG) LLM. In this assessment, completeness is measured by evaluating the presence of essential fields such as titles, URLs, and content across all extracted pages. Missing data in these fields could indicate potential gaps in documentation, incomplete records, or inaccessible content, which may impact the retrievability and quality of generated responses in a RAG-based AI system.

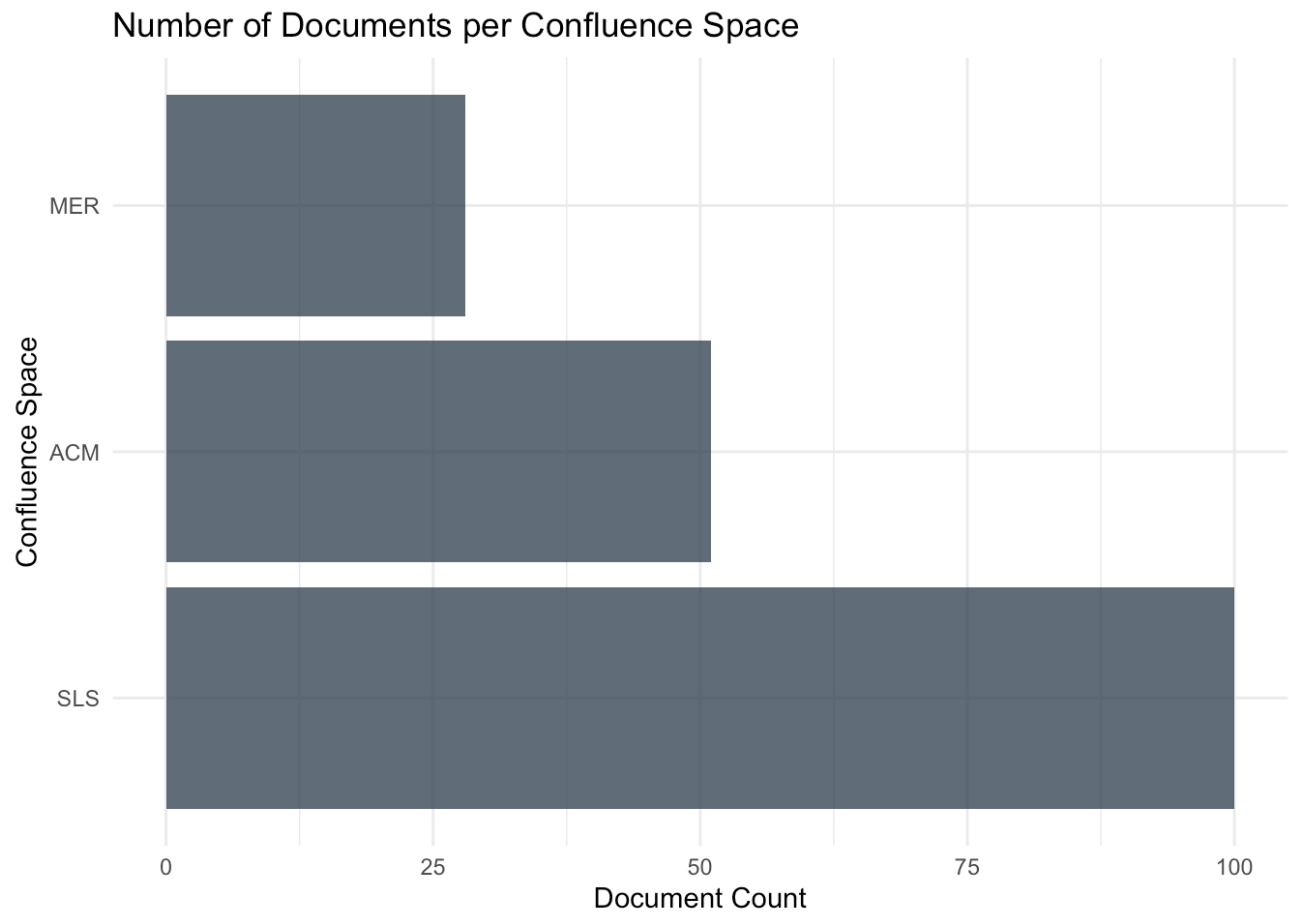
A high level of missing content, particularly in the "Content" field, suggests the presence of either empty placeholders, broken links, or restricted-access pages, which could lead to inefficient retrieval and hallucinations in the generative model. Missing titles or URLs may hinder proper indexing, making it

difficult to retrieve relevant knowledge efficiently. By quantifying missing data and visualizing its distribution, we can identify systemic gaps that need to be addressed—either through content enrichment, additional data pulls, or pre-processing techniques—before integrating this dataset into a semantic search and retrieval pipeline for AI-driven customer support automation.



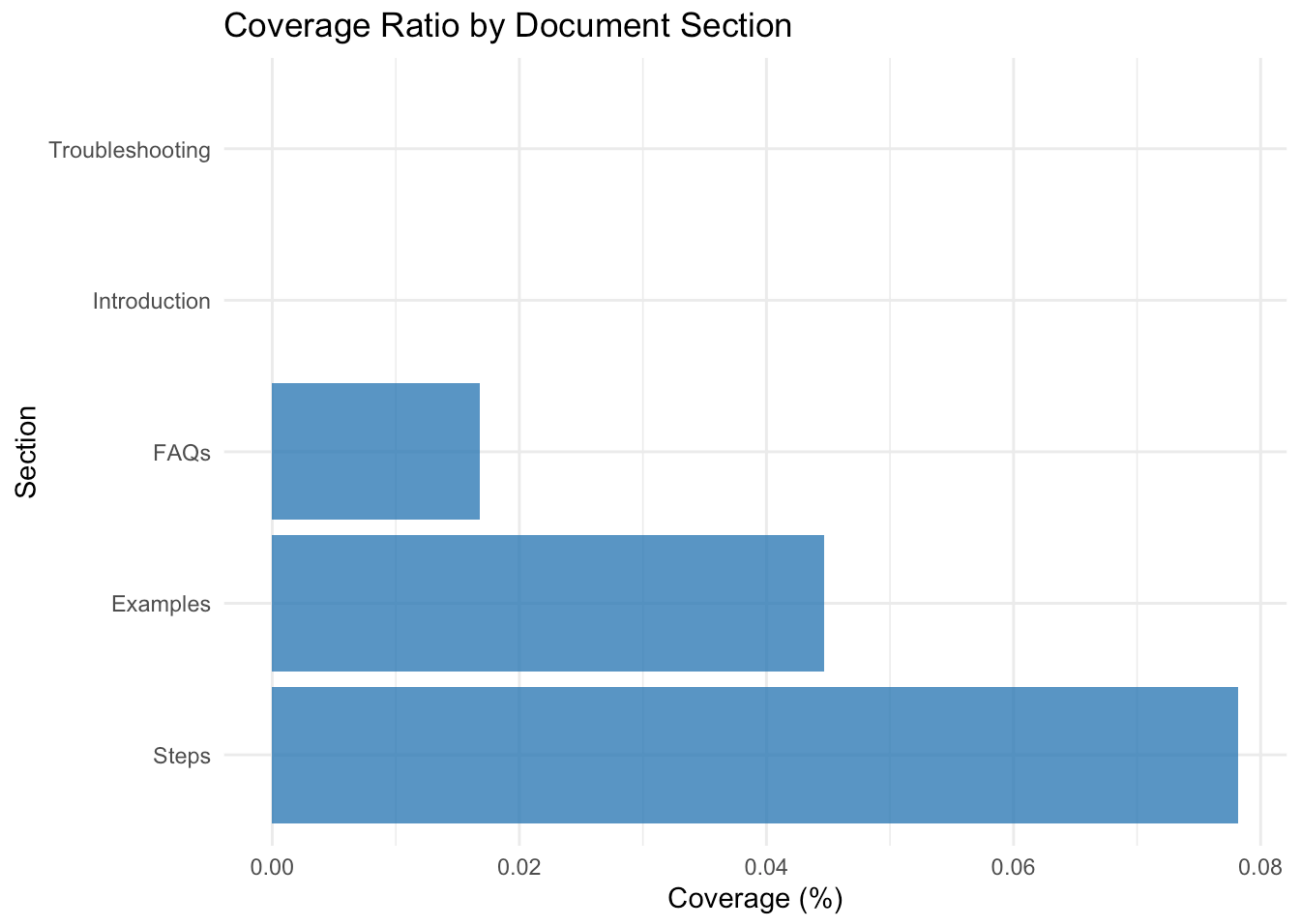
Bar Chart of Documents Per Confluence Space

*This chart provides an overview of document distribution across different Confluence spaces, helping assess whether some areas are underrepresented and require additional documentation.*



### Coverage Ratio by Document Section

*This visualization examines how frequently key structural sections (e.g., "Introduction," "Steps," "Examples") appear across documents, ensuring consistency in content formatting and completeness.*



## Microanalysis

The Microanalysis phase provides a detailed examination of the fundamental statistical properties of the Confluence repository, ensuring its readiness for retrieval and generative AI processing. Through frequency data analysis, we identify common terms and potential redundancies, while token distribution analysis evaluates the diversity and density of language usage across documents. Correlation analysis further explores the interrelationships between document attributes, revealing patterns that impact retrieval efficiency and content consistency.

Together, these insights allow us to assess documentation structure, identify biases, and optimize preprocessing strategies for effective information retrieval in a RAG-based LLM. By understanding how terms are distributed, how frequently they appear, and how different document elements interact, we can refine chunking methods, improve indexing efficiency, and enhance knowledge retrieval accuracy within GCP Vertex AI's retrieval pipeline. This ensures that the dataset is not only technically ingestible but also structured in a way that maximizes contextual relevance and generative response quality for customer service automation.

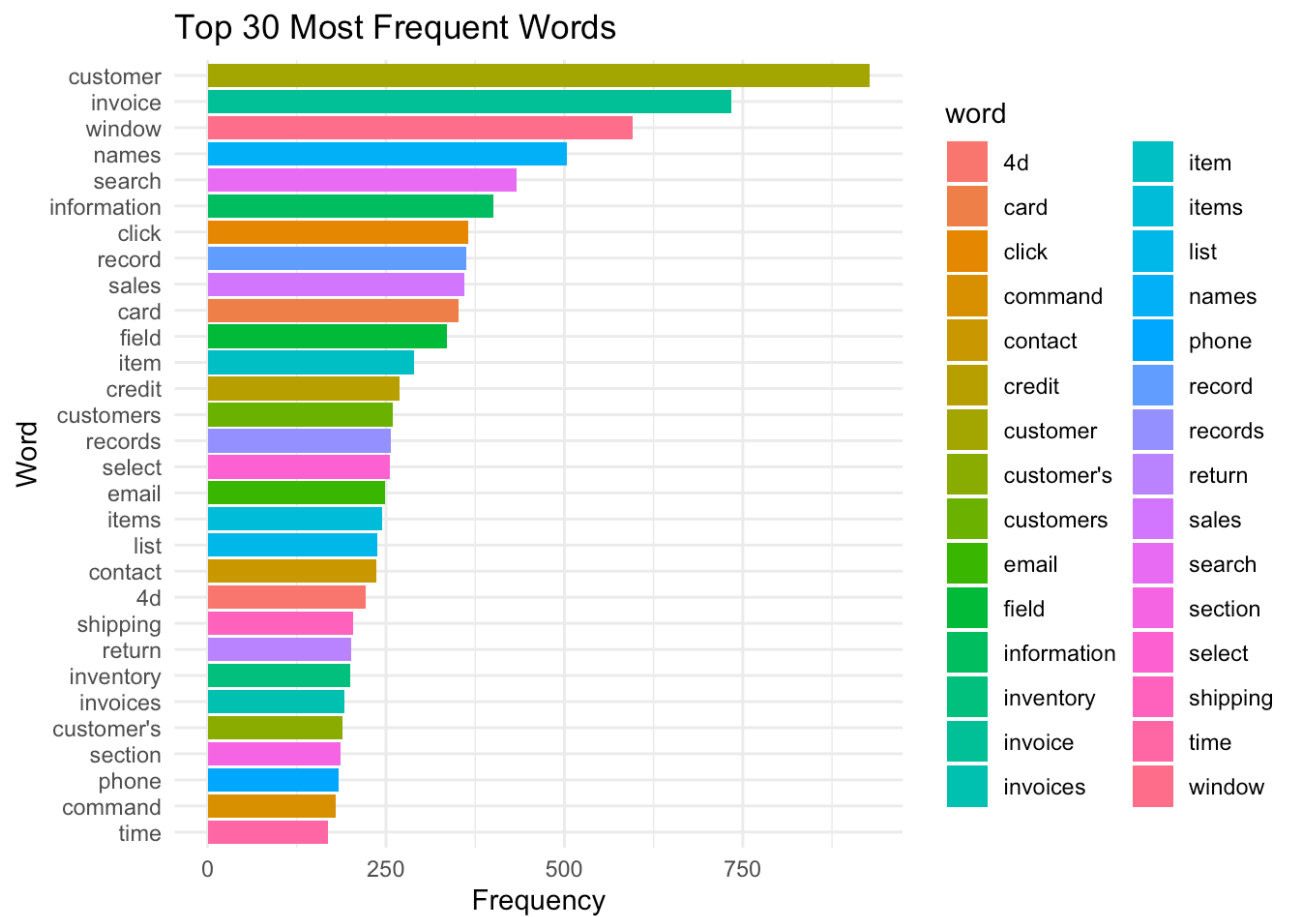
## Frequency Data

Frequency data analysis provides critical insights into the distribution of terms within the Confluence repository, helping to assess content balance, redundancy, and topical emphasis. By identifying the most

frequently occurring words, we can determine whether the dataset is well-structured and informative or if it is dominated by excessive repetition, jargon, or low-value filler text.

A healthy frequency distribution ensures that key customer service topics are well-represented, supporting effective retrieval in a RAG-powered generative AI system. Conversely, over-reliance on certain terms may indicate documentation bias, which could impact the precision and relevance of AI-generated responses. Understanding term frequency also allows for optimized search term weighting, ensuring that important but less common terms are not overlooked in retrieval. This analysis informs preprocessing decisions, enhances search relevance, and strengthens the overall knowledge retrieval strategy for Vertex AI-powered customer support automation.

|                      |                   |                 |               |              |
|----------------------|-------------------|-----------------|---------------|--------------|
| ## # A tibble: 1 × 4 |                   |                 |               |              |
| ##                   | TotalUniqueTokens | MedianFrequency | MeanFrequency | MaxFrequency |
| ##                   | <int>             | <dbl>           | <dbl>         | <int>        |
| ## 1                 | 7608              | 1               | 6.17          | 928          |

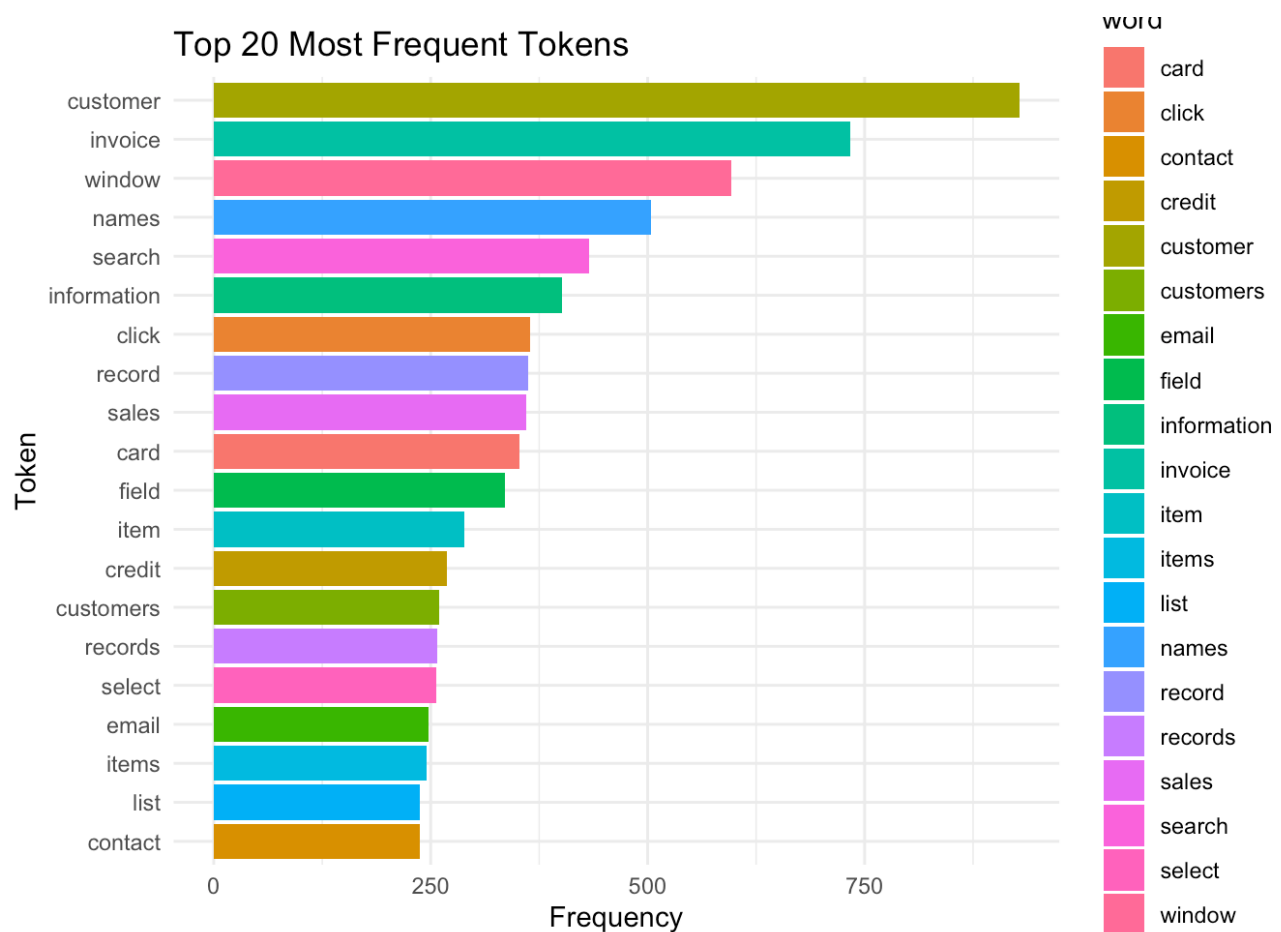


## Token Distribution Analysis

Token distribution analysis is essential for evaluating the linguistic structure and content density of the Confluence repository, ensuring it is well-suited for retrieval-augmented generation (RAG) within GCP Vertex AI. By examining the distribution of word frequencies, phrase repetition, and stopwords prevalence, we can determine whether the dataset contains rich, diverse, and meaningful information or if it is dominated by redundant, low-value content.

A well-balanced token distribution supports efficient retrieval by ensuring that high-value terms are well-represented across documents, enabling the LLM to generate contextually relevant responses. If certain tokens appear disproportionately, it may indicate terminology bias, content duplication, or documentation inconsistencies that could affect model performance. By analyzing token spread and frequency variations, we can develop better chunking strategies, optimize vector search indexing, and ensure that relevant information is surfaced effectively, ultimately improving customer support interactions powered by Vertex AI.

```
## # A tibble: 20 × 2
##   word      n
##   <chr>    <int>
## 1 customer  928
## 2 invoice  734
## 3 window   596
## 4 names    504
## 5 search   433
## 6 information 401
## 7 click    365
## 8 record   362
## 9 sales    360
## 10 card    352
## 11 field   335
## 12 item    289
## 13 credit  269
## 14 customers 260
## 15 records 257
## 16 select  256
## 17 email   248
## 18 items   245
## 19 list    238
## 20 contact 237
```

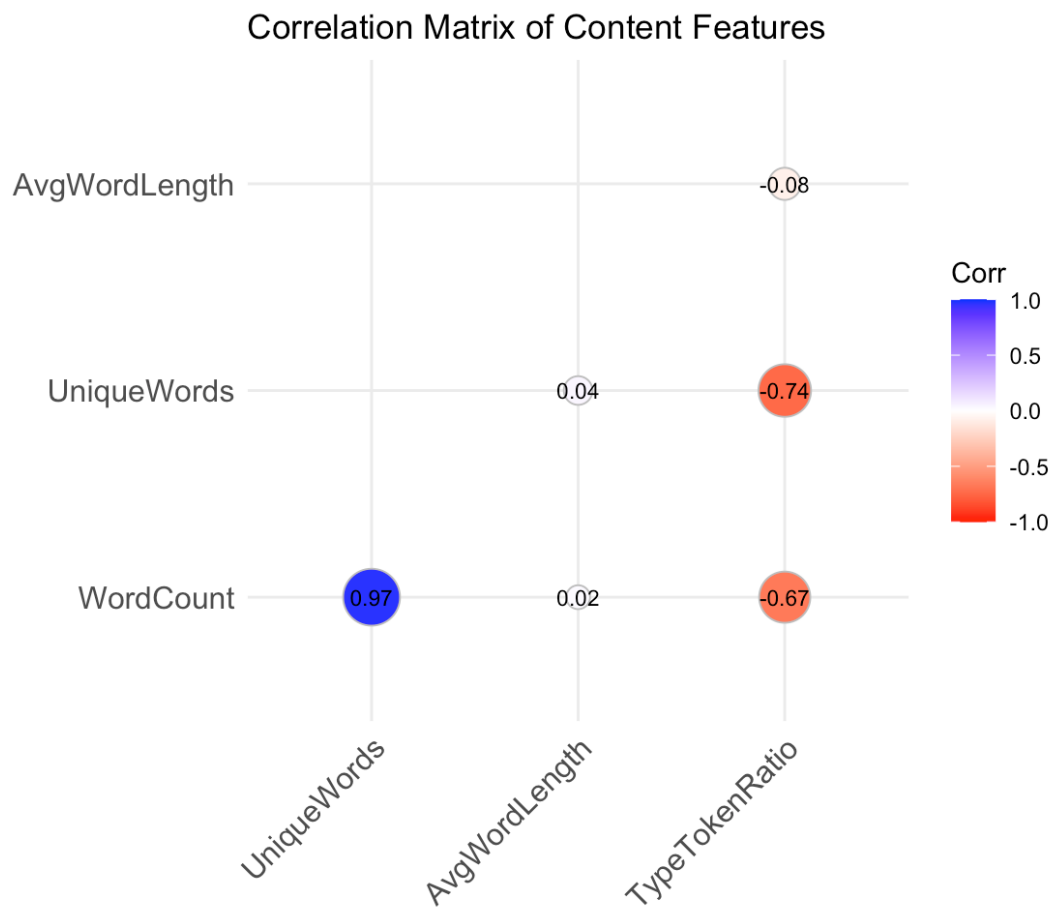


## Correlation

Correlation analysis plays a crucial role in understanding the interdependencies between document attributes, which directly impact retrieval quality and the effectiveness of a generative RAG LLM. By examining relationships between word count, missing content, title length, and URL structure, we can uncover patterns that inform data preprocessing and retrieval optimization.

For instance, if shorter documents correlate with higher missing content, it may indicate incomplete or low-value pages that could negatively affect retrieval results. Conversely, a strong correlation between title length and word count might suggest that titles can serve as effective query anchors for search optimization. Understanding these relationships allows us to refine chunking strategies, improve metadata indexing, and enhance knowledge retrieval efficiency, ensuring that the dataset is well-prepared for ingestion into GCP Vertex AI's retrieval pipeline for customer support automation.

```
## Rows: 179 Columns: 4
## — Column specification
##
## Delimiter: ","
## chr (4): Space, Title, URL, Content
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```



## Mesoanalysis

The Mesoanalysis phase provides a deeper linguistic and structural assessment of the Confluence repository, focusing on lexical variety, richness, and data quality to evaluate its suitability for ingestion into GCP Vertex AI for a Retrieval-Augmented Generation (RAG) system. By analyzing Mean Word Frequency and Type-Token Ratio, we measure the diversity of language usage, which directly impacts how effectively the AI system can retrieve and generate contextually relevant responses. The Hapax Legomena analysis further refines this understanding by identifying unique, infrequent terms that may be crucial for retrieving specialized knowledge while ensuring that essential yet underrepresented concepts are not lost.

Additionally, Data Quality and Variety Analysis helps us detect redundancy, gaps, or inconsistencies in the dataset, ensuring that the AI model processes well-balanced, information-rich content rather than overly repetitive or sparse documentation. High lexical entropy and Type-Token Ratios indicate a well-distributed knowledge base, while anomalies in word count variability may suggest incomplete or excessively verbose content that requires preprocessing adjustments.

By integrating these insights, Mesoanalysis enables us to refine document chunking strategies, retrieval indexing, and AI response calibration, ensuring that GCP Vertex AI ingests a dataset optimized for high-quality retrieval and precise, knowledge-driven customer interactions.

## Lexical Variety: Mean Word Frequency and Type-Token

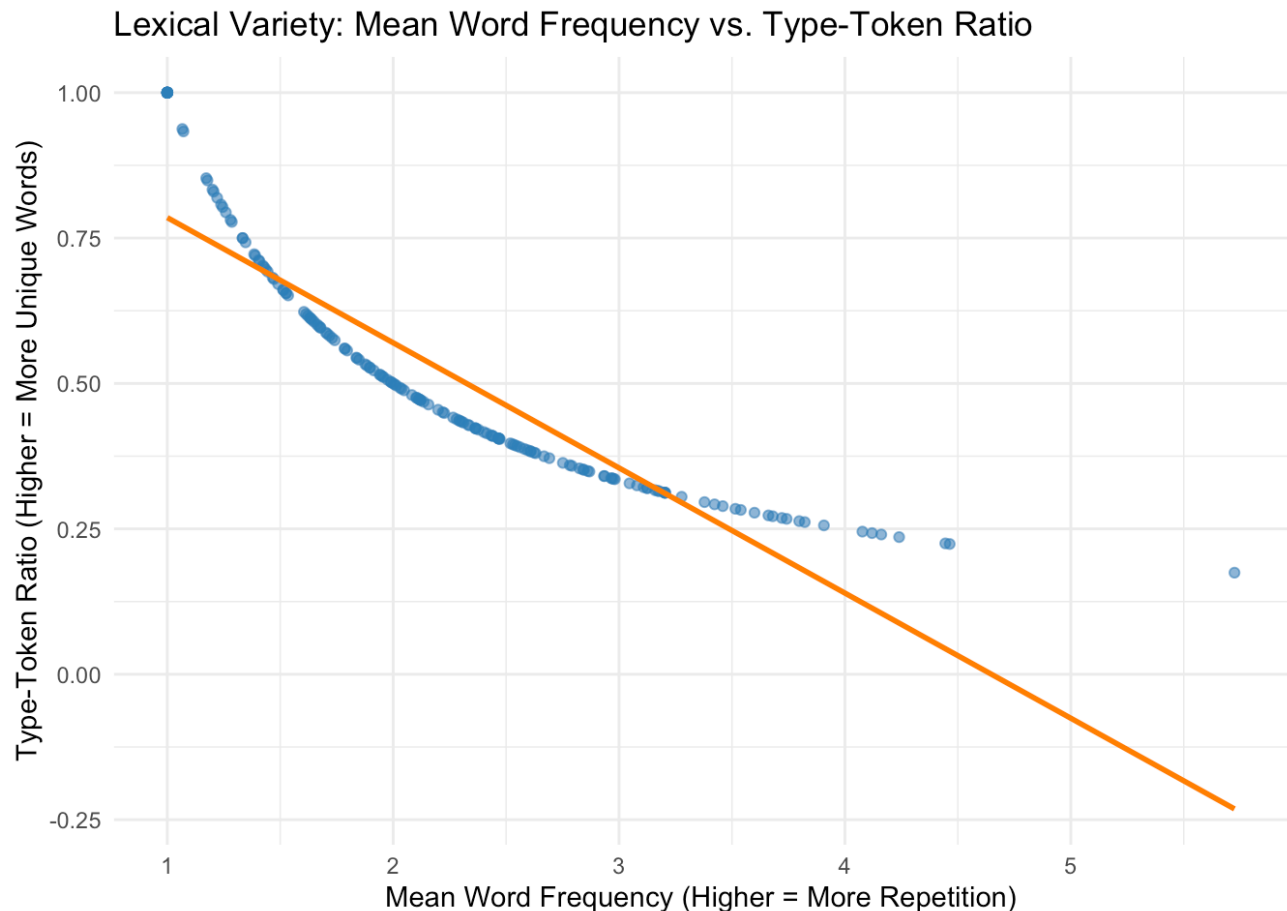
# Ratio

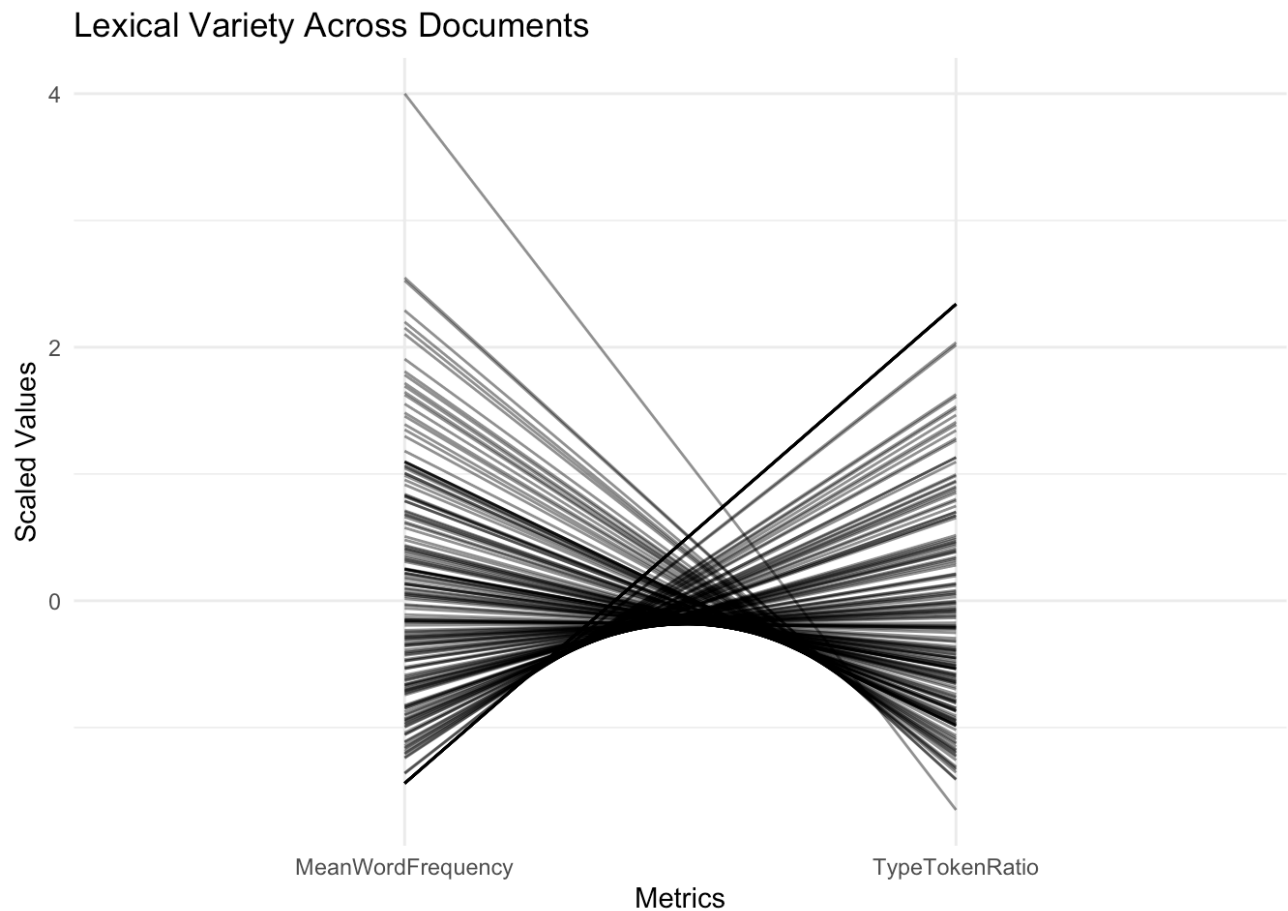
Lexical variety, measured through Mean Word Frequency and Type-Token Ratio (TTR), is a crucial indicator of content diversity and informational richness within the Confluence repository. A high TTR suggests a broad and varied vocabulary, indicating comprehensive and well-documented knowledge, while a low TTR may signal repetitive content, limited phrasing, or redundant documentation, which can negatively impact retrieval quality in a RAG-based AI system.

Mean Word Frequency helps identify whether the documentation is heavily skewed towards certain repeated terms or if the language use is balanced and diverse. Overuse of specific words may introduce bias in AI retrieval, leading to over-represented topics and weaker coverage of less frequent but important concepts. Understanding these lexical patterns allows for preprocessing improvements, such as stopwords handling, synonym mapping, and chunking strategies, ensuring that the dataset is linguistically diverse and contextually rich—critical for effective retrieval and response generation in GCP Vertex AI.

```
## # A tibble: 1 × 3
##   AvgUniqueWords AvgTotalWords AvgTypeTokenRatio
##           <dbl>         <dbl>           <dbl>
## 1           202.           583.             0.516
```

```
## `geom_smooth()` using formula = 'y ~ x'
```





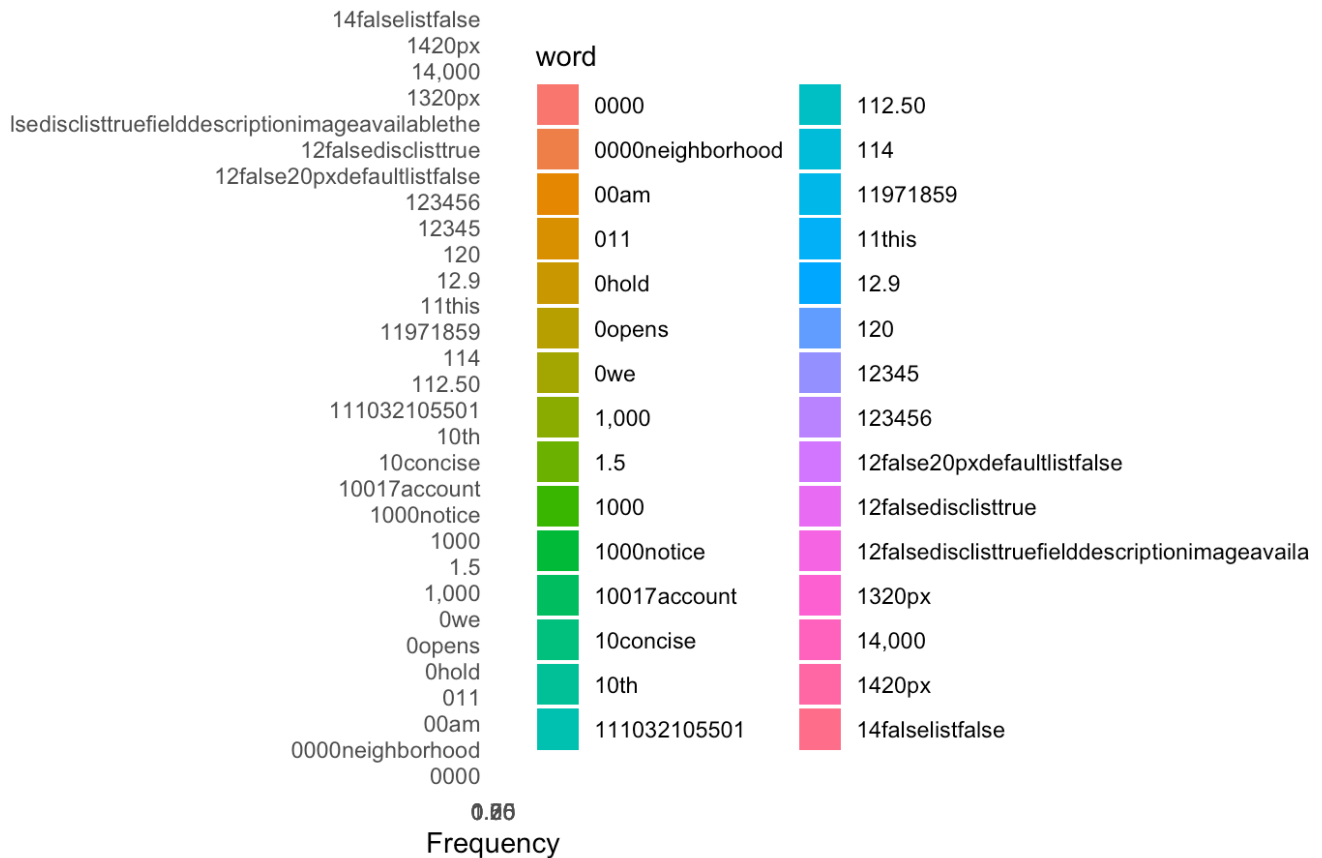
## Lexical Richness: Hapax

Hapax Legomena, or words that appear only once in a dataset, serve as a crucial indicator of lexical richness and diversity within the Confluence repository. A high Hapax count suggests that the dataset contains a wide range of unique terms, which is beneficial for knowledge retrieval and contextual AI responses. Conversely, a low Hapax count may indicate that the documentation is highly repetitive, potentially leading to redundant or biased AI outputs in a RAG-powered LLM.

By analyzing Hapax Legomena frequency and distribution, we can assess whether the documentation covers a broad spectrum of customer service scenarios or if certain concepts and terminologies are underrepresented. This insight informs preprocessing strategies, synonym expansion, and retrieval optimization, ensuring that critical but less common terms remain discoverable within GCP Vertex AI's knowledge retrieval system.

```
## # A tibble: 1 × 3
##   TotalUniqueWords HapaxCount HapaxRatio
##           <int>      <int>      <dbl>
## 1             8193        4365        0.533
```

## Top 30 Hapax Legomena



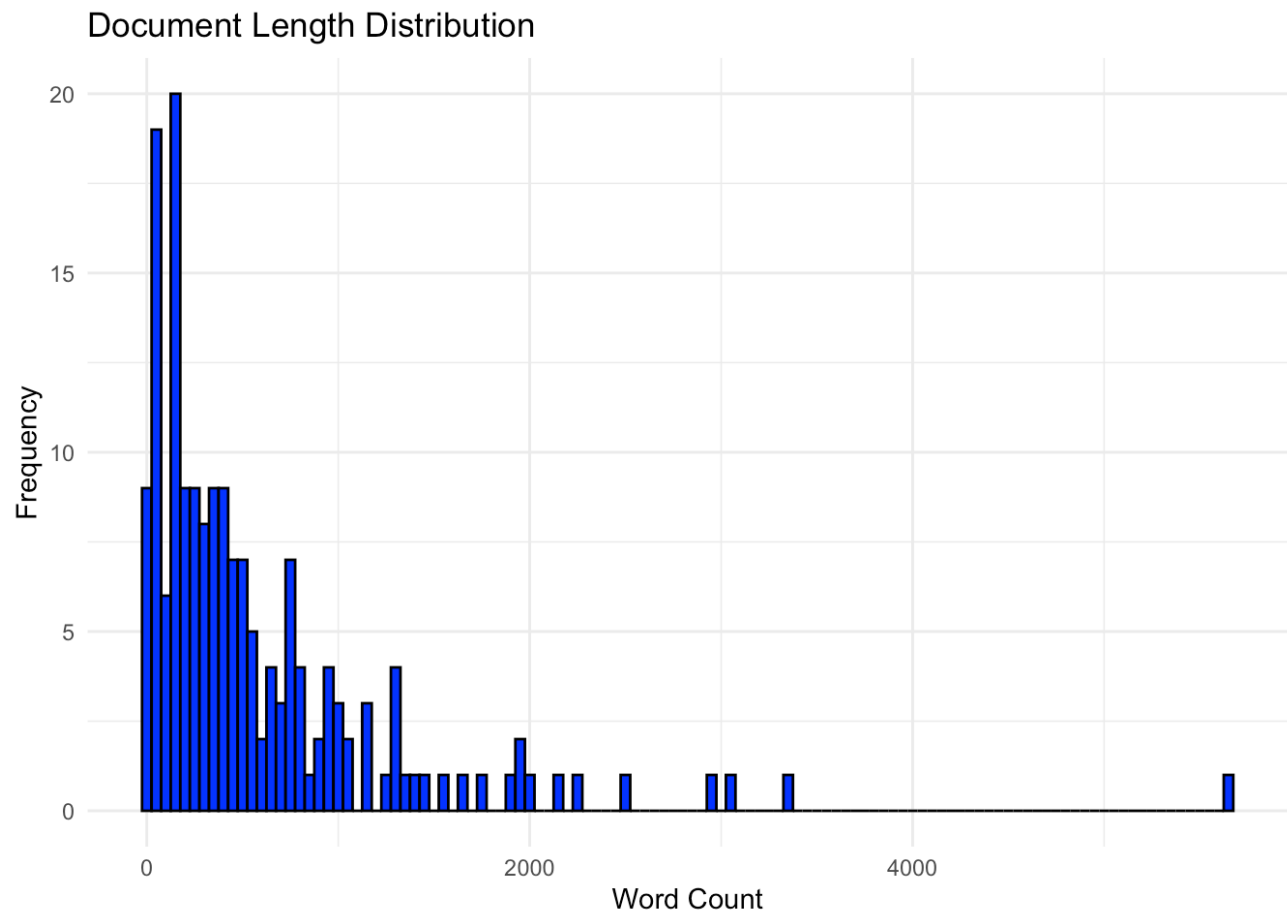
## Data Quality and Variety Analysis

Data quality and variety analysis ensures that the Confluence repository contains well-structured, diverse, and non-redundant content, which is essential for effective knowledge retrieval and AI-driven response generation. High-quality data should exhibit balanced document lengths, minimal redundancy, and a wide range of vocabulary, allowing the RAG-based LLM to generate informative, contextually rich responses rather than repeating the same limited phrases.

A low lexical variety or high redundancy may indicate that certain documents provide duplicative or overly simplified information, which can negatively impact the retrieval efficiency of the AI system. Conversely, a high Type-Token Ratio and lexical entropy suggest greater linguistic diversity, improving the model's ability to retrieve precise, relevant information across a broad set of queries. By assessing content distribution, document length variability, and term diversity, we can ensure that the dataset is optimized for ingestion into GCP Vertex AI, leading to more accurate and contextually aware customer service automation.

```
## # A tibble: 1 × 6
##   TotalPages AvgWordCount MaxWordCount MinWordCount AvgTypeTokenRatio
##   <int>      <dbl>      <int>      <int>      <dbl>
## 1      179      584.      5658        1      0.566
## # i 1 more variable: LexicalEntropy <int>
```

```
## Warning: Removed 5 rows containing non-finite outside the scale range
## (`stat_bin()`).
```



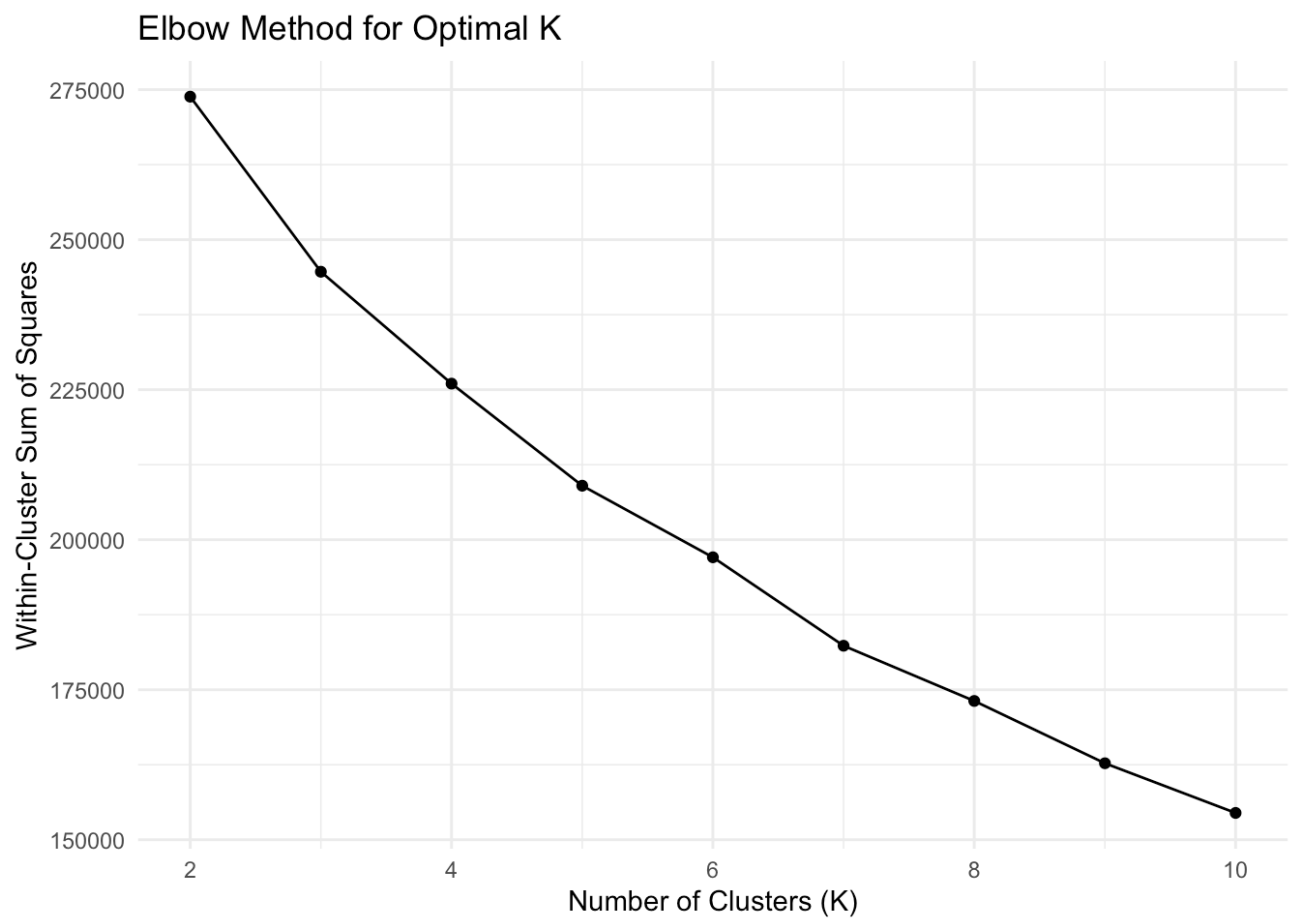
## Macroanalysis

Macroanalysis examines the high-level structure and thematic organization of the Confluence dataset, focusing on clustering, classification, and topic modeling. By identifying natural groupings of documents, we can assess whether similar content is well-organized or fragmented, which impacts AI retrieval efficiency. Classification analysis helps determine if documents can be reliably categorized, ensuring that AI-driven responses are contextually appropriate. Topic modeling further reveals hidden themes within the dataset, helping structure knowledge for semantic search and retrieval-augmented generation (RAG). These insights ensure that GCP Vertex AI can efficiently surface relevant information while minimizing redundancy and retrieval errors in a CSR Generative LLM.

## Clustering

Clustering analysis is essential for identifying natural groupings of documents within the Confluence repository, allowing us to assess how well-organized and thematically structured the data is for retrieval in a RAG-powered LLM. By applying TF-IDF vectorization and K-Means clustering, we can determine whether similar documents are grouped together, which helps in optimizing search retrieval, document chunking, and indexing strategies for GCP Vertex AI.

A well-clustered dataset ensures that related knowledge is retrieved cohesively, allowing the AI system to provide more contextually relevant responses to user queries. Conversely, poorly defined clusters may indicate inconsistencies in documentation structure, redundant information, or missing linkages between related topics. Understanding these clusters allows us to refine preprocessing strategies, ensuring that the retrieval system efficiently surfaces the most relevant information for customer service automation.

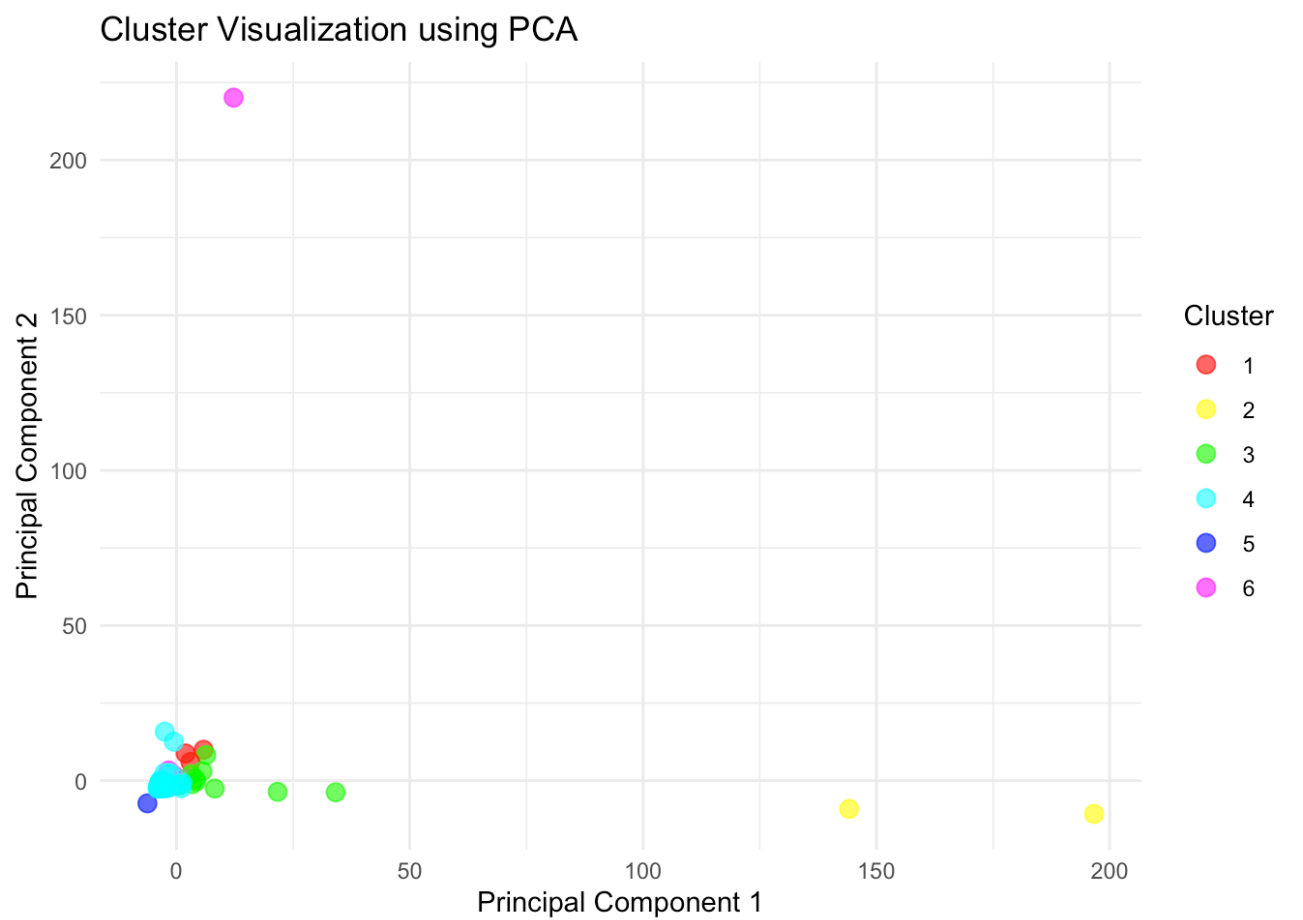


```
## # A tibble: 6 × 2
##   Title                                Cluster
##   <chr>                                <fct>
## 1 AC Manual                            2
## 2 General Duties                       5
## 3 Texting With Customers (SMS)         1
## 4 Office Hours                         2
## 5 Phone Calls                          3
## 6 Business Plans                       1
```

Visualizing document clusters provides an intuitive way to assess data organization and retrieval effectiveness. By using Principal Component Analysis (PCA), we reduce the high-dimensional document feature space into a 2D representation, making it easier to observe how well the clustering algorithm has separated different topics.

If the clusters are well-separated in the PCA plot, it suggests that distinct topic areas exist within the documentation, supporting structured, contextually relevant AI retrieval. However, if the clusters are overlapping, it may indicate that documents cover multiple themes or require further preprocessing to

improve retrieval accuracy. By analyzing this visualization, we can refine data preprocessing steps, ensuring that the AI model is trained on clean, well-organized, and logically segmented knowledge before integration into GCP Vertex AI’s retrieval pipeline.



## Classification

Classification analysis plays a key role in structuring the Confluence knowledge base by categorizing documents into predefined or learned topic groups. This process enhances retrieval efficiency in a RAG-based AI system by allowing the model to associate content with relevant subject areas, ensuring accurate, context-aware responses in customer support applications.

By leveraging Naïve Bayes classification, we can determine how well documents align with existing labels (if available) or inferred clusters. A high classification accuracy indicates that the dataset has distinct, well-defined categories, making it ideal for structured retrieval and AI-assisted navigation. Conversely, a low accuracy score may suggest overlapping content, ambiguous labels, or the need for further preprocessing.

In the context of GCP Vertex AI, effective classification ensures that the retrieval pipeline efficiently directs queries to the most relevant content, reducing retrieval errors and improving response accuracy. By refining classification techniques and ensuring well-separated categories, we can enhance AI-driven knowledge retrieval for customer service automation.

The confusion matrix provides a clear summary of the classification model’s performance on the CSR

dataset. Each cell in the matrix represents the number of documents where the actual class (Reference) corresponds to the predicted class (Prediction). Here’s what it highlights:

- **Diagonal Cells:** These represent correctly classified documents. A higher concentration in these cells indicates good model performance.
- **Off-Diagonal Cells:** These show misclassifications, where the predicted class does not match the actual class. Patterns here can help identify systematic errors in the model.
- **Color Gradient:** The intensity of the color reflects the frequency of classifications in each cell, providing an intuitive visual cue for the matrix’s distribution.

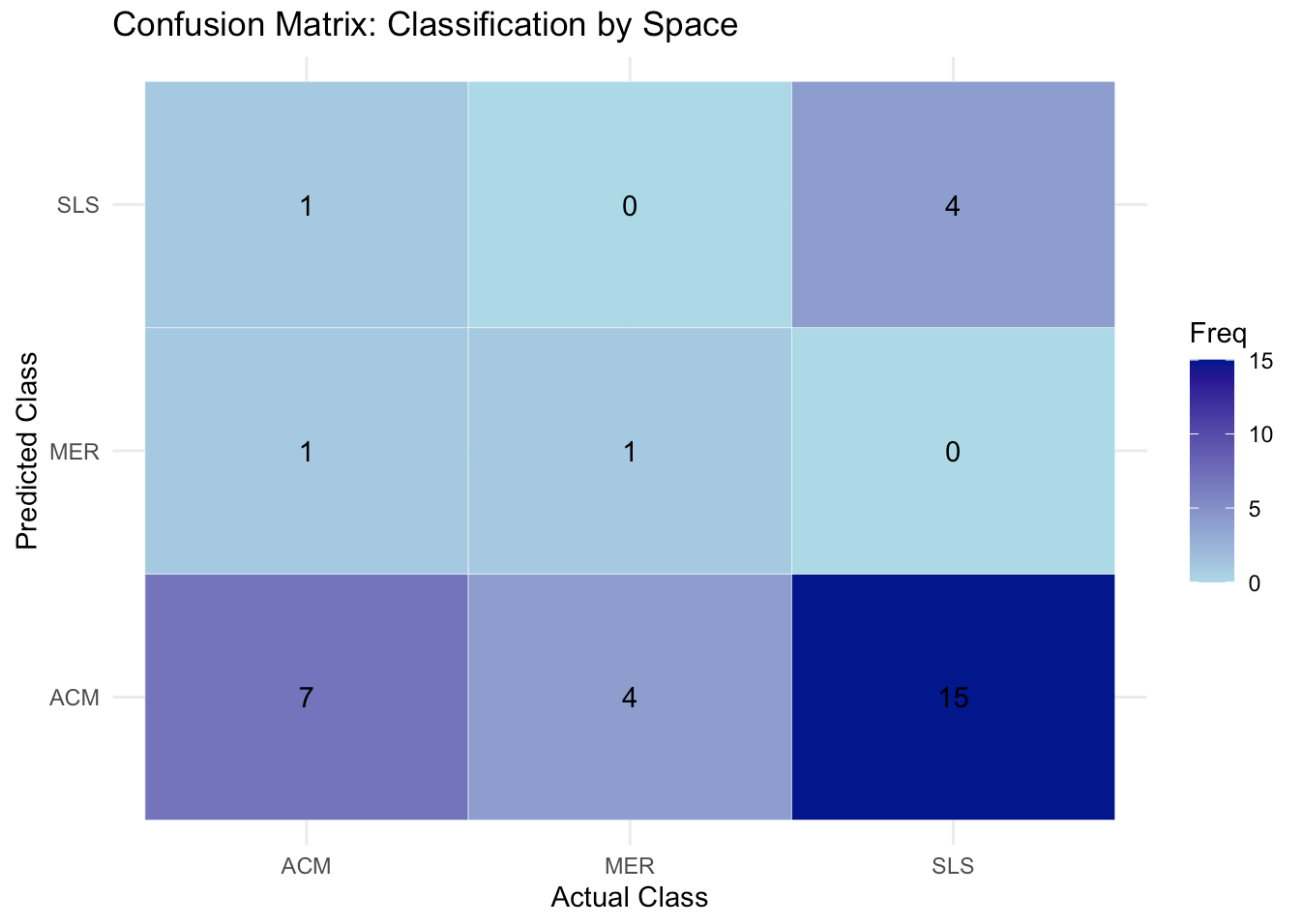
This analysis demonstrates how well the data supports initial classification efforts, helping

understand data readiness for a Generative RAG LLM.

```
## Duplicate Titles in CSR Data:
## # A tibble: 2 × 2
##   Title      n
##   <chr>    <int>
## 1 Canceling Invoices    2
## 2 Shipping             2
```

```
## CSR Data Rows: 172
```

```
## DTM Rows: 172
```



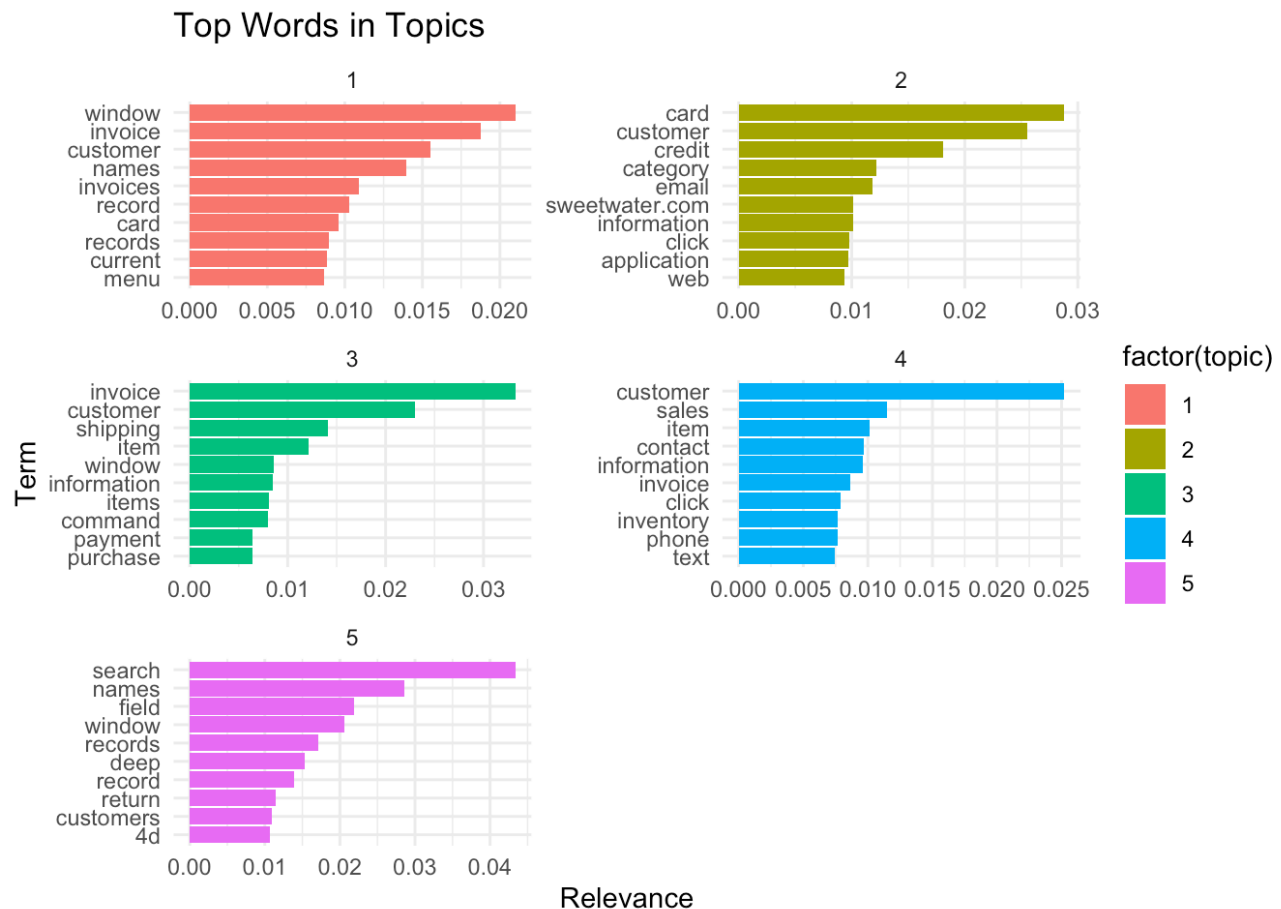
# Topic Modeling

Topic modeling is a critical step in understanding the latent themes embedded within the Confluence repository, helping to structure unorganized knowledge into distinct categories. By applying Latent Dirichlet Allocation (LDA), we can uncover underlying patterns in the documentation, identifying key topics that frequently emerge across multiple pages.

A well-structured topic model ensures that retrieval-augmented generation (RAG) models can efficiently surface relevant information, improving the accuracy and contextual relevance of AI-generated responses. If topics are well-separated, it indicates that the dataset is logically segmented, making it easier for the AI system to retrieve specific knowledge without unnecessary overlap. However, if topics are highly mixed, it may suggest a need for further content refinement or hierarchical structuring.

In the context of GCP Vertex AI, topic modeling helps to optimize search and retrieval processes, ensuring that queries pull from the most relevant documents while minimizing noise. By refining topic segmentation, we can enhance semantic search, AI-driven assistance, and document classification, leading to more precise and efficient customer support automation.

```
## # A tibble: 50 × 3
## # Groups:   topic [5]
##   topic term      beta
##   <int> <chr>    <dbl>
## 1     1 1 window  0.0210
## 2     1 1 invoice 0.0188
## 3     1 1 customer 0.0155
## 4     1 1 names   0.0140
## 5     1 1 invoices 0.0109
## 6     1 1 record  0.0103
## 7     1 1 card    0.00959
## 8     1 1 records 0.00898
## 9     1 1 current 0.00882
## 10    1 1 menu    0.00869
## # i 40 more rows
```



# Takeaways

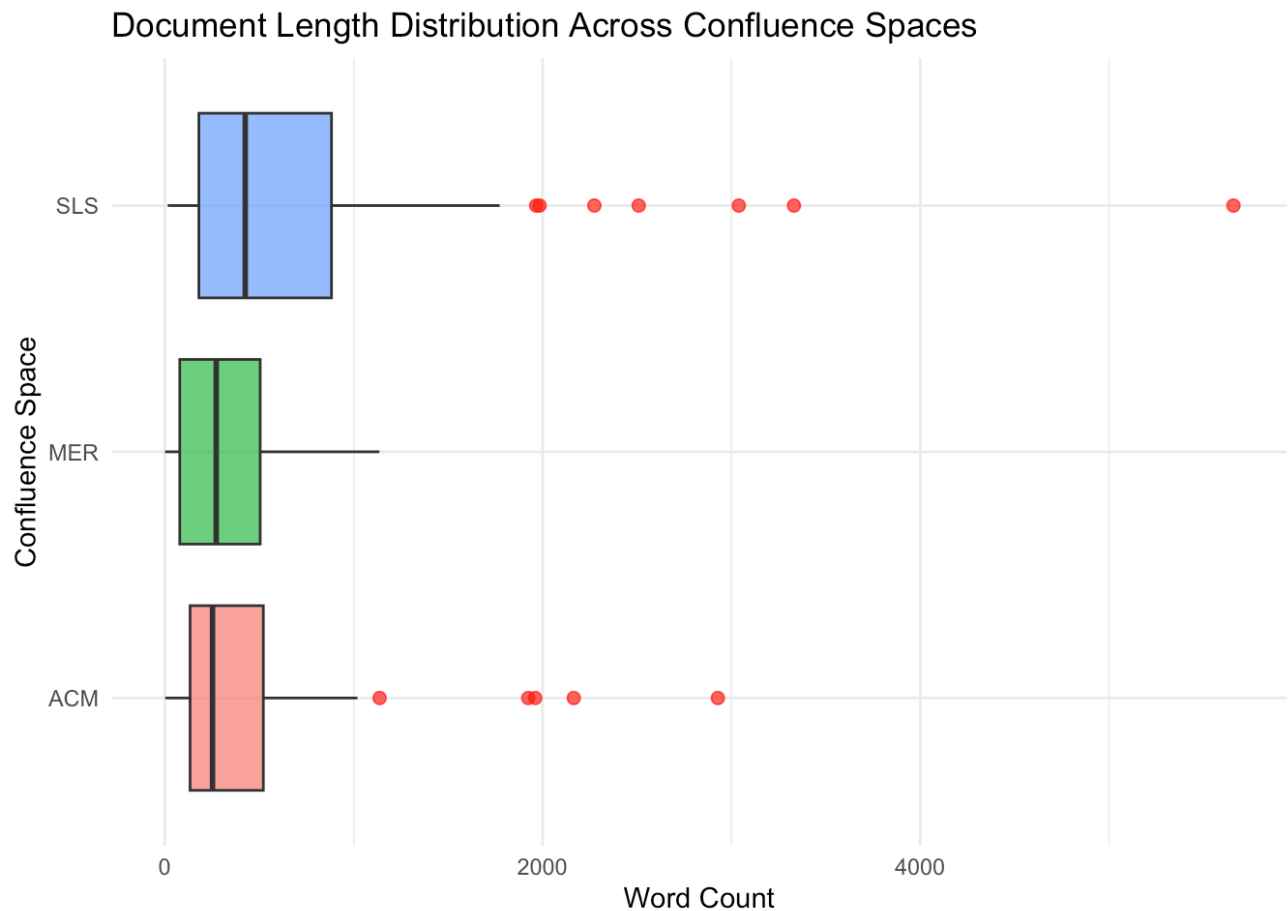
## Takeaways: Assessing Confluence Data Readiness for GCP Vertex AI and a CSR Generative LLM

The comprehensive analysis of the Confluence repository has provided a multi-faceted assessment of the dataset's structure, quality, and suitability for ingestion into GCP Vertex AI as part of a Retrieval-Augmented Generation (RAG) Large Language Model (LLM) tailored for Customer Service Representatives (CSRs). The findings indicate that the data is well-structured but requires specific preprocessing optimizations to enhance retrieval accuracy, topic alignment, and generative response quality.

### Structural Integrity and Completeness (Dimensionality Analysis)

The dimensionality assessment focused on statistical composition and data completeness, revealing key insights about document length distribution, missing data, and content density. We found that:

- The dataset contains a wide range of document sizes, with some being highly detailed while others are fragmentary or incomplete.
- Missing titles, URLs, and content gaps need to be addressed before deploying an AI-driven retrieval system to avoid surfacing incomplete responses.
- The word count distribution visualization highlighted variability in content richness across documents, reinforcing the need for content standardization.

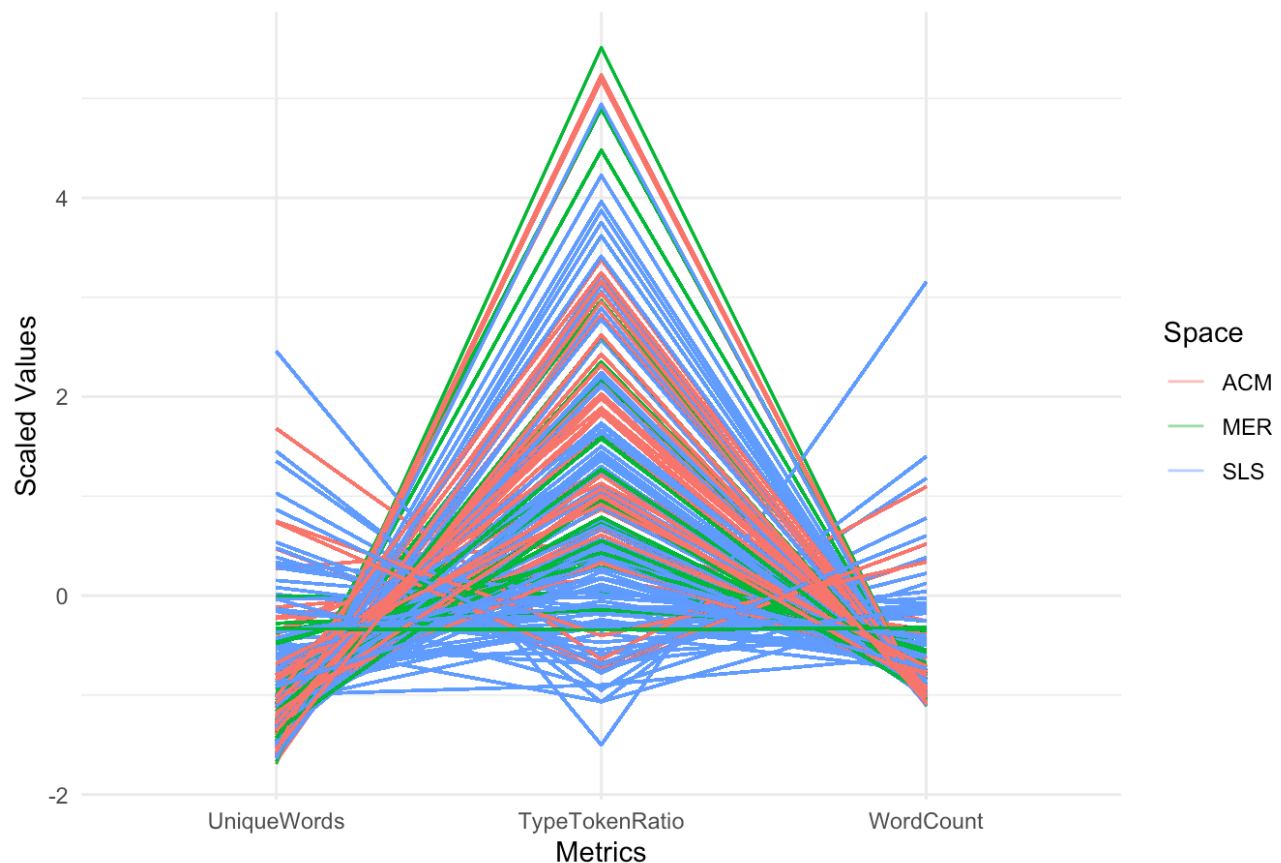


### Lexical and Linguistic Complexity (Mesoanalysis)

The Mesoanalysis provided deeper insight into word frequency, vocabulary diversity, and overall linguistic richness.

- Token Distribution Analysis revealed that some terms dominate the corpus, which could create biases in the AI model's response generation.
- Lexical Variety and Type-Token Ratio (TTR) confirmed that certain documents are rich in unique vocabulary, ensuring that specialized topics are well-represented.
- Hapax Legomena (words appearing only once) provided a deeper look into niche topics within the dataset.
- Data Quality and Variety Analysis confirmed content redundancy in certain sections, which may lead to overfitting in the AI's retrieval process.

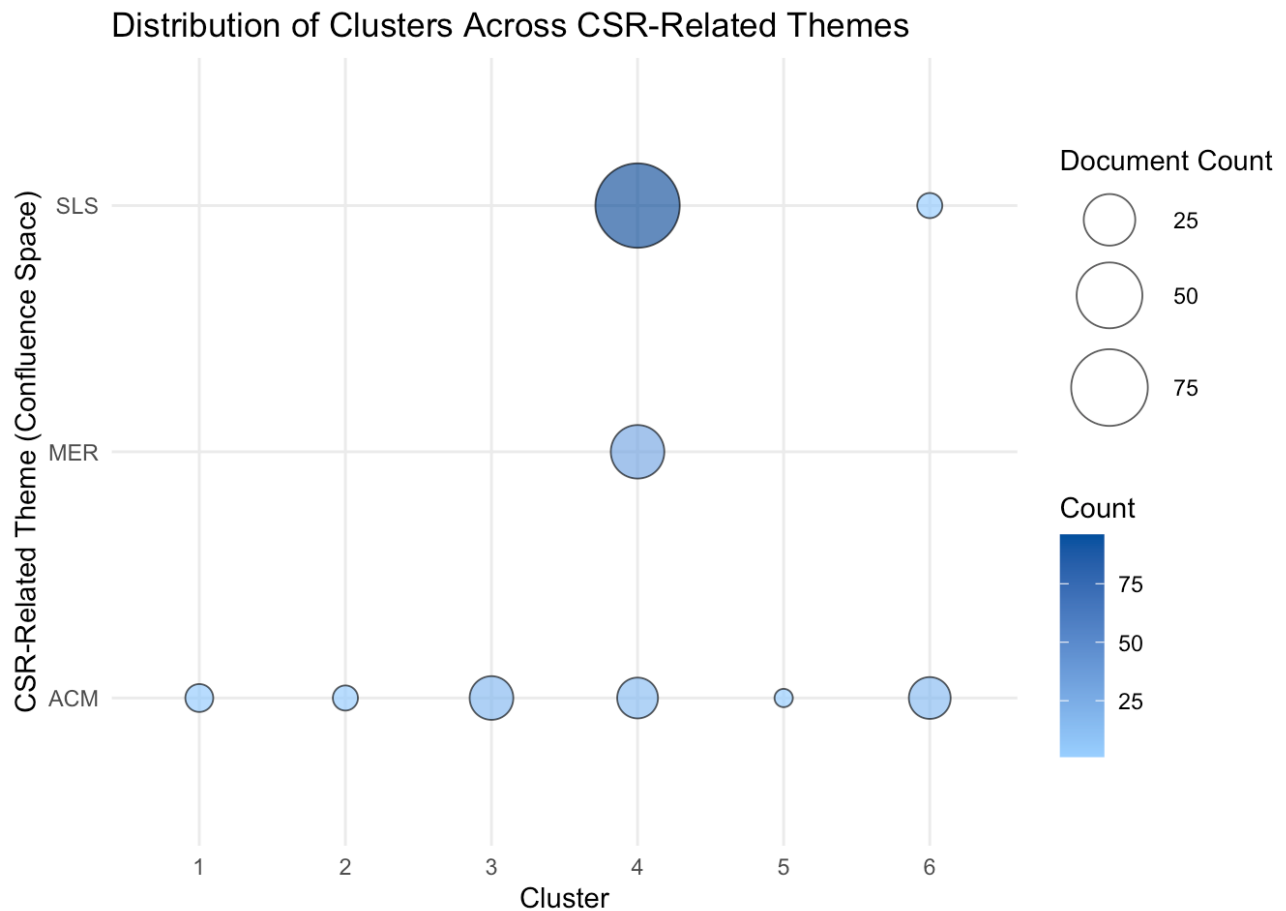
## Lexical Diversity Across Confluence Spaces



### Thematic and Organizational Structure (Macroanalysis)

At a broader level, Macroanalysis focused on how documents cluster together, how they can be classified, and what topics emerge.

- Clustering Analysis (K-Means) uncovered natural groupings of documents, suggesting that some topics are well-organized, while others have significant overlap, indicating potential content redundancies.
- Classification Analysis (Naïve Bayes) demonstrated that certain topics are well-separated within the dataset, meaning the AI model can effectively categorize responses based on CSR queries.
- Topic Modeling (LDA) extracted dominant themes across the dataset, confirming that key topics such as product support, customer inquiries, and troubleshooting workflows are well-covered but require more granular sub-topic structuring.



The bubble chart visualizes the distribution of document clusters across CSR-related themes, helping to identify areas where content is well-structured vs. where topic overlap exists. Larger bubbles indicate high-density clusters, suggesting that certain topics are well-defined and consistently documented, while smaller or widely dispersed bubbles may indicate fragmented or overlapping themes. This insight helps assess whether CSR documentation is logically organized for retrieval and whether clustering refinement is needed to improve AI-assisted knowledge retrieval in GCP Vertex AI.

### Why GCP Vertex AI Is the Right Approach for a CSR Generative LLM

The findings from this analysis strongly support GCP Vertex AI as the ideal platform for deploying a CSR-focused RAG-based Generative LLM. Key reasons include:

- **Optimized Retrieval Pipelines** – GCP Vertex AI’s embedding-based search will effectively rank and retrieve relevant CSR documents, mitigating concerns around overlapping content and redundant retrieval.
- **Fine-Tuned Model Training with Quality Data** – By leveraging data preprocessing techniques, we can improve document segmentation and token prioritization, allowing the model to generate clear, accurate, and contextually rich responses.
- **Scalability and Performance for CSR Workflows** – With Vertex AI’s scalable infrastructure, the system can efficiently serve real-time customer inquiries, improving response times and CSR support efficiency.
- **Integration with Enterprise Knowledge Graphs** – Vertex AI’s semantic search and knowledge

graph capabilities will enhance document linking, ensuring that related topics are surfaced dynamically to provide holistic responses to CSR queries.

## Next Steps and Additional Visualizations

To further refine data readiness for deployment, the following next steps are recommended:

1. **Preprocessing Enhancements** – Improve missing content handling, duplicate removal, and document segmentation to optimize retrieval.
2. **Fine-Tuning Generative AI Responses** – Utilize fine-tuned embeddings to ensure precise context matching between queries and retrieved content.
3. **Dynamic Document Structuring** – Implement hierarchical topic indexing to improve semantic relevance.



## Final Takeaway: GCP Vertex AI Enables a More Intelligent, Scalable CSR LLM

This analysis confirms that your CSR Confluence repository contains a wealth of structured and semi-structured knowledge, making it a strong candidate for integration into GCP Vertex AI's RAG pipeline. However, targeted preprocessing and content alignment will significantly enhance retrieval performance. With optimized embeddings, clustering refinement, and topic structuring, a highly effective CSR Generative AI system can be deployed, delivering accurate, context-aware responses that improve customer support efficiency.